UK Maths Trust

Junior Mathematical Challenge

Solutions 2026

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For reasons of space, these solutions are necessarily brief.

There are more in-depth, extended solutions available on the UKMT website, which include some exercises for further investigation:
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1. C The results of the five calculations are, respectively, 26, 14, 46, 34 and 32.
So $20 + 26$ gives the largest result.

2. D The lengths of the perimeters of the five figures are, respectively, 10, 10, 10, 8 and 10.
So figure D is the odd one out.

3. B $\dfrac{4-3-2-1}{8-7-6-5} = \dfrac{-2}{-10} = \dfrac{1}{5}$.

4. E Let the number which Barry is thinking of be x.

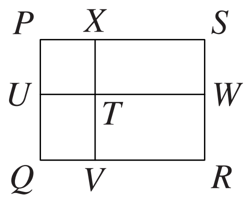
Then $\dfrac{x}{2} + \dfrac{x}{3} + \dfrac{x}{4} = 52$. So $\dfrac{6x + 4x + 3x}{12} = 52$, that is $\dfrac{13x}{12} = 52$.
Hence $x = \dfrac{12 \times 52}{13} = 12 \times 4 = 48$.

5. D As the Salisbury Cathedral Clock strikes once every hour, it strikes 24 times every day. Therefore, in the 70 years since 1956, the number of times it has struck is approximately $24 \times 365 \times 70 \approx 24 \times 350 \times 70 \approx 24 \times 25\,000 = 600\,000$.

6. C As zero cannot be the leading digit in a number, the three-digit numbers made only from 0s and 1s are 100, 101, 110 and 111. Their sum is 422. The two-digit numbers made only from 2s and 3s are 22, 23, 32 and 33. Their sum is 110. So, on Wednesday, Ada's final addition was $422 + 110$, which gave her the result of 532.

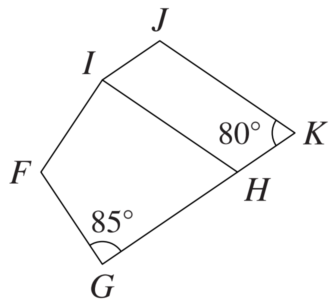
7. A All primes greater than 2 are odd. The sum of five odd numbers is also odd. However, 2026 is even and therefore one of the five primes is 2 as this is the only even prime. As 2 is also the smallest prime, this means that there is exactly one possibility for the smallest prime in Sam's list. There are many lists of five primes which sum to 2026. One example is 2, 19, 31, 983, 991.

8. **C** The area of square $PUTX$ is $9\,\text{cm}^2$, so its side-length is 3 cm. Therefore each side of rectangle $TVRW$ is 3 cm shorter than the corresponding side of rectangle $PQRS$. So its perimeter is 4×3 cm less than 30 cm, that is 18 cm.



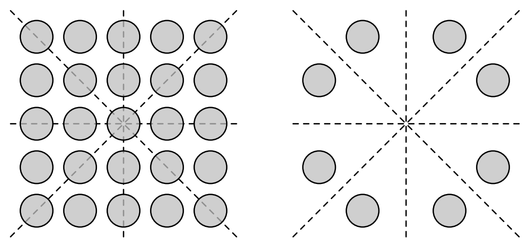
9. **D** Carl counts as follows: 1, 2, 4, 7, 8, 11, 13, 14, 16, 17, 19, 22, 23, 26, 28, ... So, after 15 items, Carl has reached 28.

10. **B** As IH and JK are parallel and G, H and K lie on a straight line, $\angle GHI = 80°$ (corresponding angles). Also, as $FGHI$ is a kite in which $IF = FG$ and $GH = HI$, $\angle FIH = \angle FGH = 85°$. Therefore, as the sum of the interior angles of a quadrilateral is $360°$, $\angle IFG = (360 - 80 - 2 \times 85)° = 110°$.



11. **A** The sum of the nine numbers to place in the grid is $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9$, that is 45. So it is impossible for the sum of the three numbers in each row to be even, as three even numbers have an even sum but 45 is odd.

12. **D** The diagram shows that the array of biscuits has exactly four lines of symmetry. John may eat all of the biscuits, one by one, along any of those lines of symmetry and the array will still be symmetric about that line while he does so. When he has eaten all of the biscuits along that line, the array will be symmetric about the line at right angles to the eaten line.

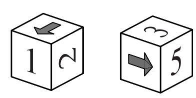


So, provided that he takes the lines of symmetry in turn, John may eat all 17 of the biscuits which lie along those lines. During this process, it is not possible to remove any of the eight biscuits shown in the diagram on the right without losing whatever lines of symmetry currently exist since these symmetries all match the eight biscuits in pairs. So the maximum number of biscuits which John can eat is 17.

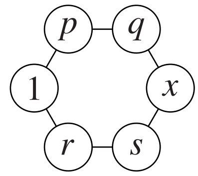
13. **E** The digits q and r are not necessarily different, so when $p = 2$, the maximum value of 'pq' + 'pr' is $29 + 29$, that is 58. When $p = 4$, the minimum value of 'pq' + 'pr' is $40 + 40$, that is 80. Therefore, as 'pq' + 'pr' = 75, we can deduce that $p = 3$. So $q + r = 75 - 2 \times 30$, that is 15. Hence $p + q + r = 3 + 15 = 18$.
(*Note that we cannot determine the values of q and r as they could be 7 and 8 in either order or 6 and 9 in either order, but in each case* $p + q + r = 18$*.*)

14. **E** Let the length of RS be l cm. The sum of the perimeters of triangles RPS and RSQ is equal to the perimeter of triangle PQR plus twice the length of RS. Hence $44 + 60 = 80 + 2l$.
So $l = (44 + 60 - 80) \div 2$, that is 12. Therefore the length of RS is 12 cm.

15. **D** The diagram shows two views of the assembled cube. As can be seen, the arrows point to 1 and to 5. So the required sum is 6.



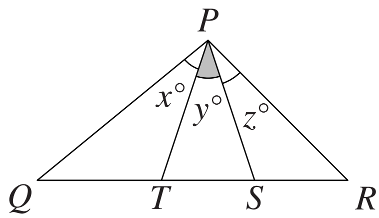
16. A Note that p, q, r and s are as shown on the diagram. The sum of the numbers in each pair of adjacent circles is a prime number, so we may deduce that p and r are both even. Hence q and s are both odd. So they are 3 and 5 in either order. This leaves 2, 4 and 6 for the values of p, r and x, though not necessarily in that order. However, x cannot be 4 or 6, as $4 + 5$ and $3 + 6$ both equal 9, which is not prime. However, $2 + 3$ and $2 + 5$ both result in primes, so $x = 2$. One possible solution is $p = 4, q = 3, r = 6, s = 5, x = 2$. It is left to the reader to discover if there are any other solutions.



17. B Clearly, Hamlin must start by writing down '1'. He may then write down a '0' or a '1'. If he starts '10' then the third number he writes down must be '1'. There are now exactly two different ways in which he can complete his list, namely '101100' and '101010'. However, if Hamlin starts '11', then the third '1' may be in exactly three different positions, so the list may then be completed in three further ways, namely '111000', '110100' and '110010'. So Hamlin can write down the three 1s and three 0s in exactly five different ways.

18. D If Amy and Bob are both telling the truth then Amy has more pencils then Dan. However, Dan claims to have more than Amy, so at most four of the children are telling the truth. If the order from the holder of the greatest number of pencils to the child with the lowest number is Eve, Amy, Bob, Cat, Dan then the first four of these children are indeed telling the truth.

19. C Let the sizes of angles QPT, TPS and SPR be $x°$, $y°$ and $z°$ respectively. Then, as $\angle QPR = 100°$, $x + y + z = 100$. [1]
Now $QP = QS$, so $\angle TSP = (x + y)°$.
Also, $RP = RT$, so $\angle PTR = (y + z)°$. Hence $\angle PTS = (y + z)°$.
The sum of the interior angles of triangle PTS is 180°.
So $y + (y + z) + (x + y) = 180$, that is $x + 3y + z = 180$. [2]



Subtracting equation [1] from equation [2] now gives $2y = 80$, so $y = 40$. Therefore $\angle TPS$ is 40°.

20. E Let the number of coffee cakes be n. Then the number of cream cakes is $\frac{3n}{2}$.

So the number of cherry cakes is $\frac{5}{4} \times \frac{3n}{2} = \frac{15n}{8}$ and the number of chocolate cakes is $\frac{7}{6} \times \frac{15n}{8} = \frac{105n}{48} = \frac{35n}{16}$. Therefore the total number of cakes is $n + \frac{3n}{2} + \frac{15n}{8} + \frac{35n}{16}$

$$= \frac{16n + 24n + 30n + 35n}{16} = \frac{105n}{16}.$$

So, as $\frac{105}{16}$ cannot be simplified, n is a multiple of 16 and the total number of cakes is a multiple of 105. Of the options given, only 420 satisfies this condition. It corresponds to $n = 64$.

21. B Let the required rectangle have dimensions, in cm, of $4x$ by $5x$ and let it be covered by k rectangles measuring 30 cm by 5 cm. Then, by considering areas, we obtain $4x \times 5x = k \times 30 \times 5$. Hence $20x^2 = 150k$. So $2x^2 = 15k$. As 15 is not a multiple of 2 and is the product of two different primes, namely 3 and 5, the minimum value of x is 15, corresponding to $k = 30$. The 30 rectangles may be arranged in a number of different ways. One way is to place 15 of the smaller rectangles side by side to create a 30 cm by 75 cm rectangle and then to place another 15 smaller rectangles below these to complete a 60 cm by 75 cm rectangle.

22. B It may be deduced that the number of frogs is a multiple of 5 and the number of snakes is multiple of 7. So the total number of creatures is the sum of a multiple of 5 and a multiple of 7.
Now $22 = 3 \times 5 + 1 \times 7$; $24 = 2 \times 5 + 2 \times 7$; $26 = 1 \times 5 + 3 \times 7$; $27 = 4 \times 5 + 1 \times 7$.
However, it is not possible for 23 to be written as the sum of a multiple of 5 and a multiple of 7. So there could not be 23 creatures in the pond.

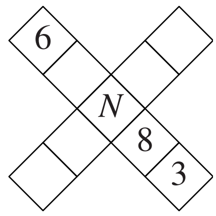
23. E Let the five numbers in ascending order be p, q, r, s, t. The mean of the first four numbers is 5.

Therefore $\dfrac{p+q+r+s}{4} = 5$. So $p+q+r+s = 4 \times 5 = 20$. [1]

The mean of the last four numbers is 7. Therefore $\dfrac{q+r+s+t}{4} = 7$.

So $q+r+s+t = 4 \times 7 = 28$. [2]
Subtracting equation [1] from equation [2]: $(q+r+s+t) - (p+q+r+s) = 28 - 20$.
Hence $t - p = 8$. So the range of the five numbers is 8.

24. A The sum of the positive integers from 1 to 9 inclusive is 45. Therefore the sum of the numbers in the two lines is $45 + N$, since N appears in both lines and needs to be counted twice. So, as the totals of the five numbers in each line are equal, N must be odd. If $N = 1$, then the equal totals are $\dfrac{45+1}{2}$, that is 23. Therefore the number in the blank square between 6 and N is $23 - (3 + 8 + 1 + 6)$, that is 5.



Note that N cannot be 3 as 3 has already been placed. If $N = 5$, then the equal totals are $\dfrac{45+5}{2}$, that is 25. So the number in the blank square between 6 and N is $25 - (3 + 8 + 5 + 6)$, that is 3. Therefore N cannot be 5 as 3 has already been placed. Working in the same way, we see that if $N = 7$, then the equal totals are 26 and the number in the blank square between 6 and N is 2. Also, if $N = 9$, then the equal totals are 27 and the number in the blank square between 6 and N is 1. Therefore the sum of the different values which the central number N can take is $1 + 7 + 9$, that is 17.

25. B We first consider the situation in which there are no ties at the finish line. The winner may be any one of four runners. Second place may then go to any of the three remaining runners and then third place to any of the two remaining runners. That then leaves exactly one runner to fill the fourth place. So if there are no ties then the number of different possible finishing orders is $4 \times 3 \times 2 \times 1$, that is 24.
We now consider the number of different finishing orders when there is exactly one tie at the finish line. Let the four runners be P, Q, R and S. For each pair of runners who tie there are six possible finishing orders — for example, if P and Q tie then these are $P/Q, R, S$; $P/Q, S, R$; $R, P/Q, S$; $S, P/Q, R$; $R, S, P/Q$; $S, R, P/Q$. Also, there are six possible pairs of runners who could tie, namely P/Q, P/R, P/S, Q/R, Q/S, R/S. So, if there is exactly one tie, then the number of possible finishing orders is 6×6, that is 36. Therefore the total number of possible finishing orders is $24 + 36$, that is 60.